Water pollution

Water Pollution

Water Pollution occurs when energy and other materials are released into the water, contaminating the quality of it for other users.

Water Pollution

- Water pollution is any chemical, biological, or physical change in water quality that has a harmful effect on living organism or makes water unsuitable for desired uses.
- It has been suggested that it is the leading worldwide cause of deaths and diseases, and that it accounts for the deaths of more than 14,000 people daily.

Water Pollutants….

- Petroleum hydrocarbons
- Plastics
- Paints and arsenics
- Heavy metals
- Sewage
- Radioactive waste
- Thermal effluents detergents
- Chloroform
- Food processing waste, (fats and grease)
- Insecticides and herbicides
- Petroleum hydrocarbons (gasoline, diesel fuel, jet fuels and fuel oil)
- Lubricants (motor oil)
- From storm water runoff

Types of water pollution

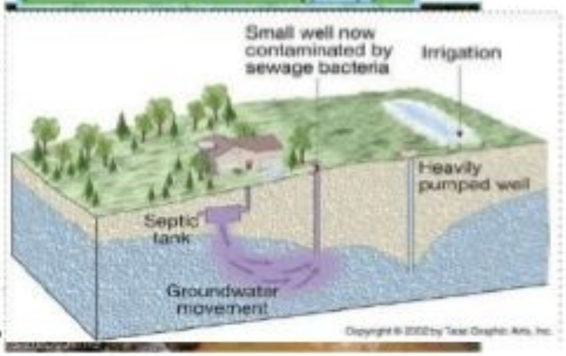
- Point source pollution

- Non point source pollution

Types of water pollution

Surface water pollution

Ground water pollution



i. Microbiological pollution

microorganisms that thrives on water fishes that can cause illness to animals and humans

ii. Oxygen Depletion pollution

microorganisms that in water and feeds on biodegradable substances

Water Stagnation

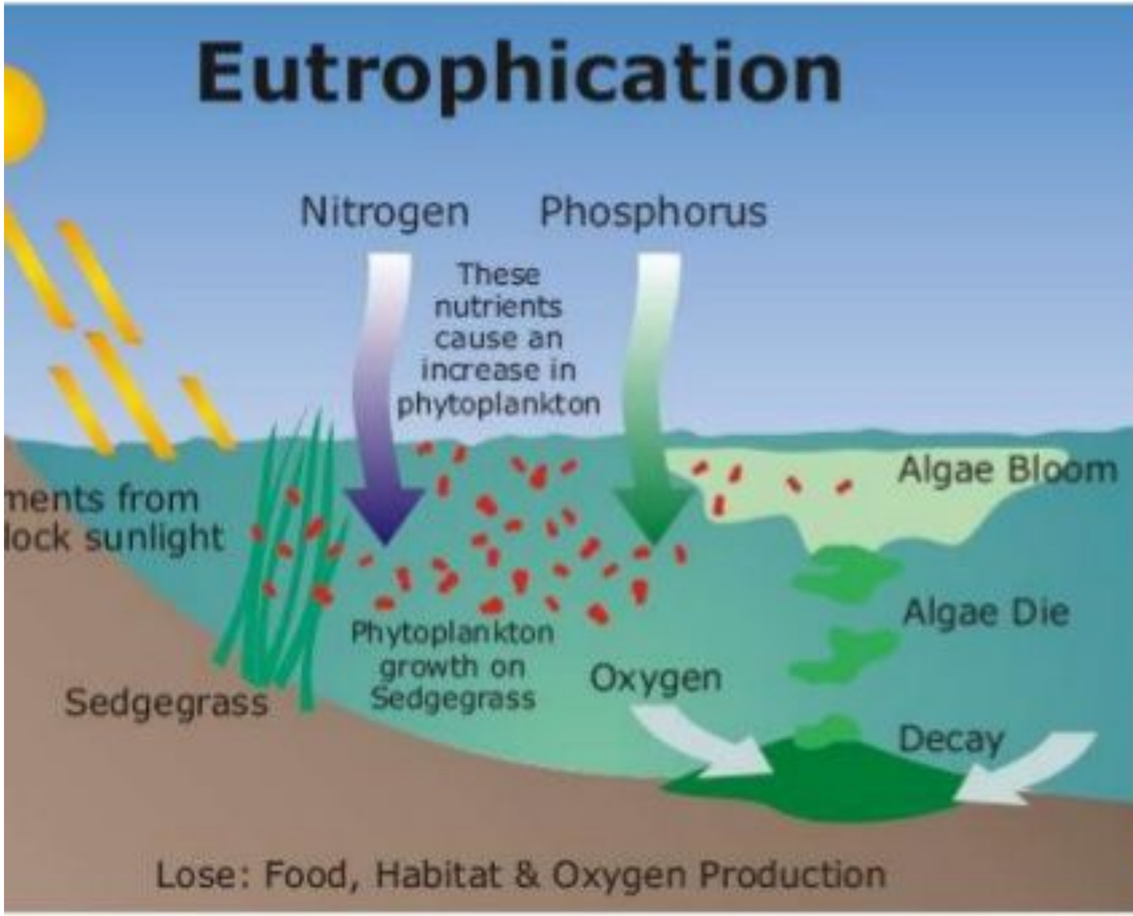
- **Occurs when water stops flowing**
  □
- **Malaria and dengue are among the main dangers of stagnant water.**

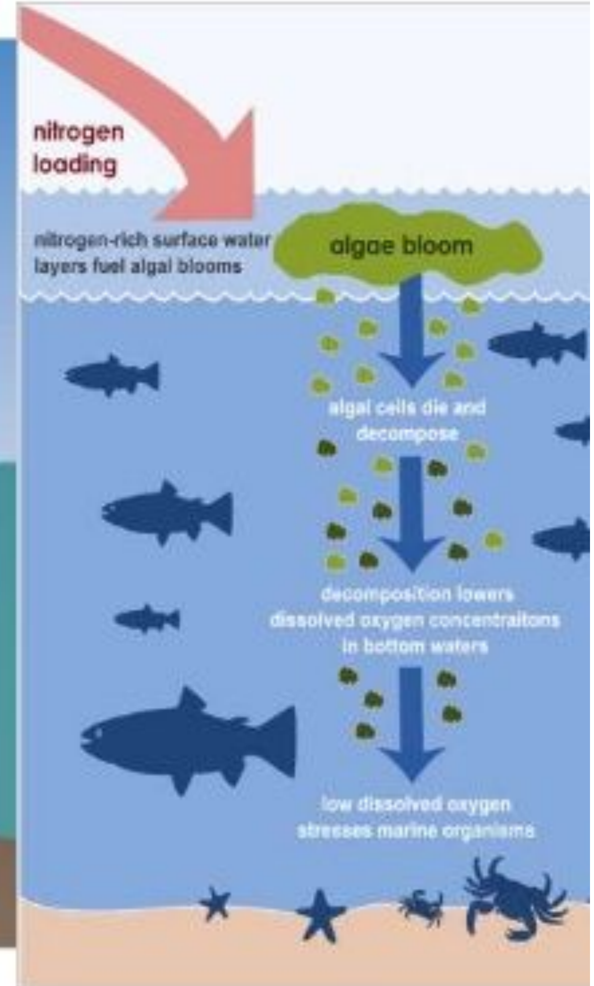
The sources of water pollution

- ❖ Municipal Waste Water
- ❖ Industrial Waste
- ❖ Inorganic Pollutants
- ❖ Organic Pollutants
- ❖ Agricultural Wastes
- ❖ Marine Pollution
- ❖ Thermal pollution
- ❖ Radioactive Pollution

Effects of Water Pollution

- Lower DO may be harmful to animals.
- **Eutrophication**.
- **Bioaccumulation.**
- **Biomagnification**.
- Minimata disease, Blue **Baby Syndrome** or **Methaemoglobinemia and** Fluorosis
- Pesticides are harmful to **aquatic life**.
- Dyes and inorganic compounds induce colour change in animals


Eutrophication
Nitrogen
Phosphorus
These nutrients cause an increase in phytoplankton
ments from
lock sunlight
Algae Bloom
Algae Die
Phytoplankton growth on Sedgegrass
Sedgegrass
Oxygen
Decay
Lose: Food, Habitat & Oxygen Production


nitrogen loading
nitrogen-rich surface water layers fuel algal blooms
algae bloom
algal cells die and decompose
decomposition lowers dissolved oxygen concentraitons in bottom waters
low dissolved oxygen stresses marine organisms

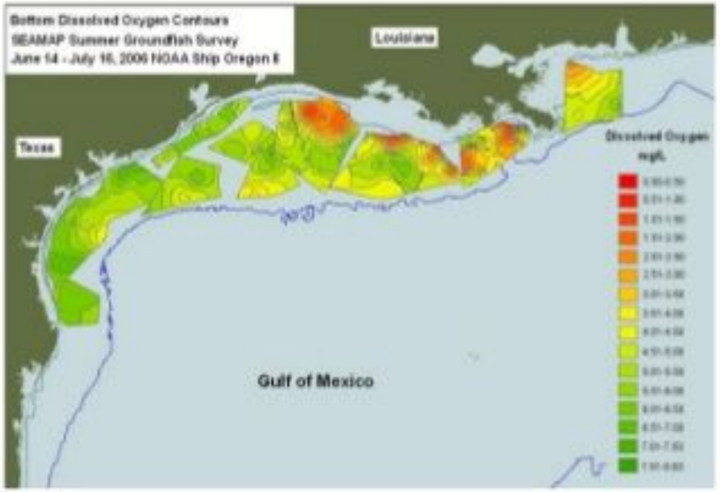
Eutrophication

- An increase in chemical nutrients — compounds containing nitrogen or phosphorus — in an ecosystem, and may occur on land or in water. However, the term is often used to mean the resultant increase in the ecosystem's primary productivity (excessive plant growth and decay), and further effects including lack of oxygen and severe reductions in water quality, fish, and other animal populations.

Eutrophication of the Potomac River.

Hypoxia

- A phenomenon that occurs in aquatic environments as dissolved oxygen (**DO**; molecular oxygen dissolved in the water) becomes reduced in concentration to a point detrimental to aquatic organisms living in the system.
- Oxygen depletion can be the result of a number of factors including natural ones, but is of most concern as a consequence of pollution and eutrophication in which plant nutrients enter a river, lake, or ocean, phytoplankton blooms are encouraged.



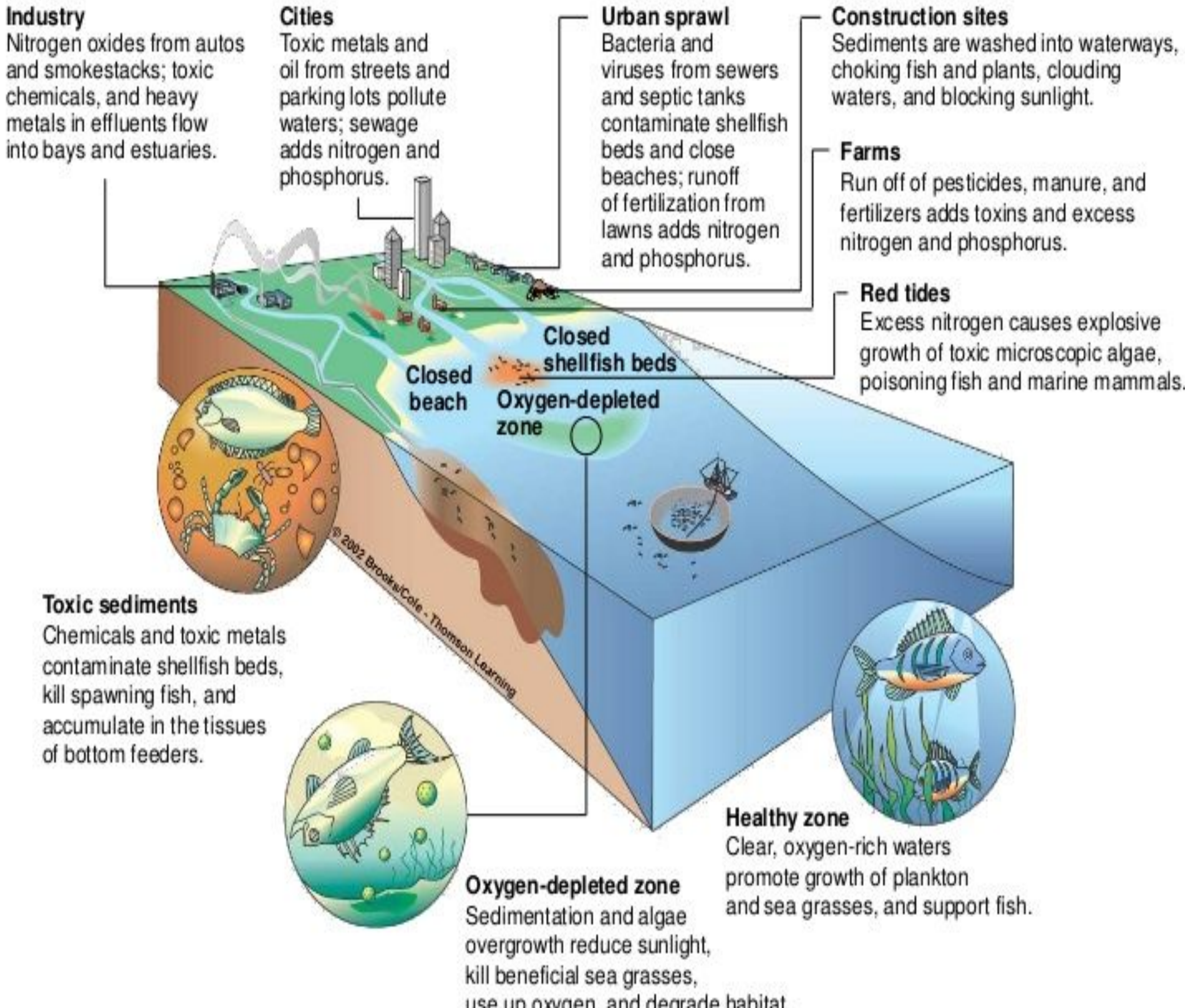
Effects on Environment

- Toxic water
- Thermal heating
- Our sources of water

Effects on human

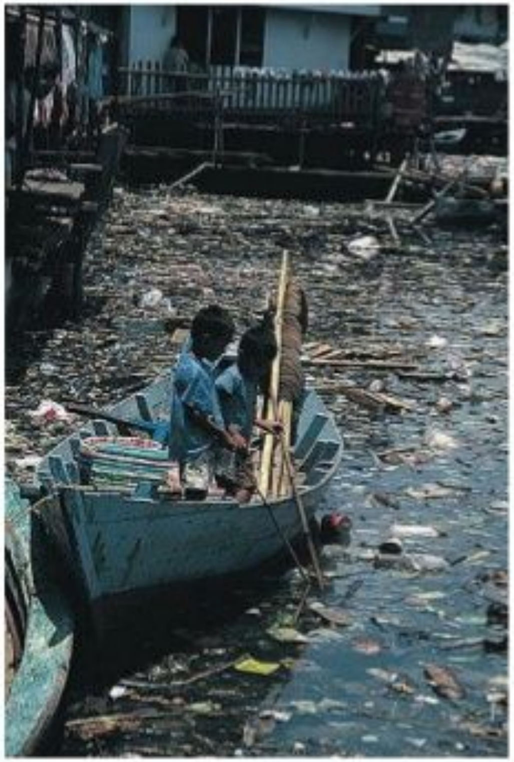
- Diseases caused by:
  - Drinking contaminated water
  - Swimming in polluted water
  - Contact with chemically polluted water

Effects on birds and animals


Industry
Nitrogen oxides from autos and smokestacks; toxic chemicals, and heavy metals in effluents flow into bays and estuaries.
Cities
Toxic metals and oil from streets and parking lots pollute waters; sewage adds nitrogen and phosphorus.
Urban sprawl
Bacteria and viruses from sewers and septic tanks contaminate shellfish beds and close beaches; runoff of fertilization from lawns adds nitrogen and phosphorus.
Construction sites
Sediments are washed into waterways, choking fish and plants, clouding waters, and blocking sunlight.
Farms
Run off of pesticides, manure, and fertilizers adds toxins and excess nitrogen and phosphorus.
Red tides
Excess nitrogen causes explosive growth of toxic microscopic algae, poisoning fish and marine mammals.
Closed shellfish beds
Closed beach
Oxygen-depleted zone
© 2002 Brooks/Cole - Thomson Learning
Toxic sediments
Chemicals and toxic metals contaminate shellfish beds, kill spawning fish, and accumulate in the tissues of bottom feeders.
Healthy zone
Clear, oxygen-rich waters promote growth of plankton and sea grasses, and support fish.
Oxygen-depleted zone
Sedimentation and algae overgrowth reduce sunlight, kill beneficial sea grasses, use up oxygen, and degrade habitat.

Marine Pollution

- Entry into the ocean of chemicals, particles, industrial, agricultural and residential waste, or the spread of invasive organisms.
- Most sources of marine pollution are land based. The pollution often comes from nonpoint sources such as agricultural runoff and wind blown debris.
- Many potentially toxic chemicals adhere to tiny particles which are then taken up by plankton and benthos animals, most of which are either deposit or filter feeders.
- Toxins are concentrated upward (biomagnification) within ocean food chains.

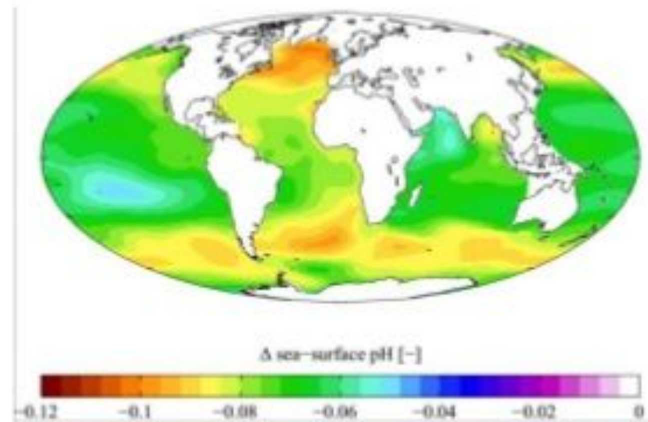


Marine Debris

- Human-created waste that has deliberately or accidentally become afloat in a lake, sea, ocean or waterway. Oceanic debris tends to accumulate at the centre of gyres and on coastlines, frequently washing aground, when it is known as beach litter.
- Plastic bags, balloons, buoys, rope, medical waste, glass bottles and plastic bottles, cigarette lighters, beverage cans, styrofoam, lost fishing line and nets, and various wastes from cruise ships and oil rigs are among the items commonly found.

Ocean Acidification

- The ongoing decrease in the pH of the Earth's oceans, caused by their uptake of anthropogenic carbon dioxide from the atmosphere.
- Human activities such as land-use changes, the combustion of fossil fuels, and the production of cement have led to a new flux of CO_2 into the atmosphere.
- Dissolving CO_2 in seawater also increases the hydrogen ion (H^+) concentration in the ocean, and thus decreases ocean pH.



Change in sea surface pH caused by anthropogenic CO_2 between the 1700s and the 1990s.

Ship Pollution

- Spills from oil tankers and chemical tankers
- Ejection of sulfur dioxide, nitrogen dioxide and carbon dioxide gases into the atmosphere from exhaust fumes.
- Discharge of cargo residues from bulk carriers can pollute ports, waterways and oceans.
- Noise pollution that disturbs natural wildlife.
- Water from ballast tanks can spread harmful algae and other invasive species.

Thermal Pollution

- The rise or fall in the temperature of a natural body of water caused by human influence.
- A common cause is the use of water as a coolant by power plants and industrial manufacturers.
- Warm water typically decreases the level of dissolved oxygen in the water . The decrease in levels of DO can harm aquatic animals.
- May also increase the metabolic rate of aquatic animals, as enzyme activity, resulting in these organisms consuming more food.

Table 2. WHO standards for drinking water

Parameter	World Health Organization
Fluoride	1.5 mg/l
Arsenic	10μg/l
Benzene	10μg/l
Boron	2.4mg/l
Cadmium	3 μg/l
Selenium	40 μg/l
Tetrachloroethene and Trichloroethene	40μg/l
Nitrate	50 mg/l
Chromium	50μg/l
Mercury	6 μg/l
Barium	700μg/l

Water Purification

- The process of removing undesirable chemical and biological contaminants from raw water.
- Designed to meet drinking, medical, pharmacology, chemical and industrial applications.
- Treatment processes also lead to the presence of some mineral nutrients: fluoride, calcium, zinc, manganese, phosphate, and sodium compounds.

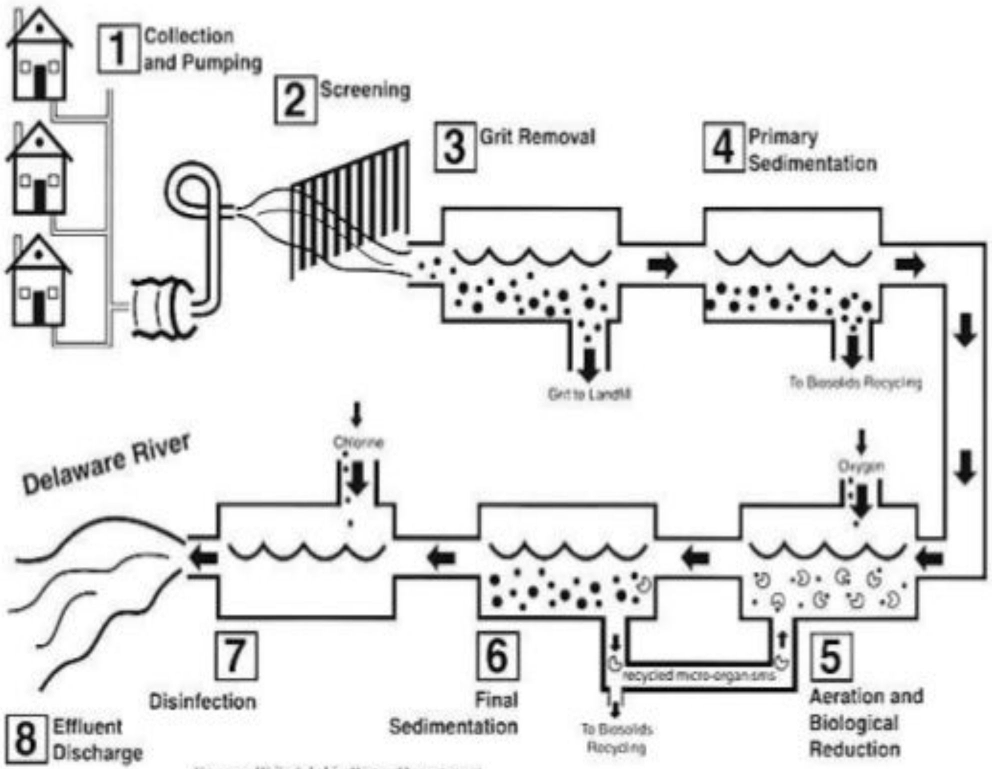
General Methods

- **Physical Process:** Filtration, Sedimentation
- **Biological Processes:** Slow Sand Filters, Activated Sludge
- **Chemical Processes:** Flocculation, Chlorination
- **Radiation:** Ultraviolet Light

Pre-Treatment

- Pumping and Containment
- Screening
- Storage
- Pre-conditioning (sodium carbontate)
- Pre-chlorination

Sewage Treatment Process

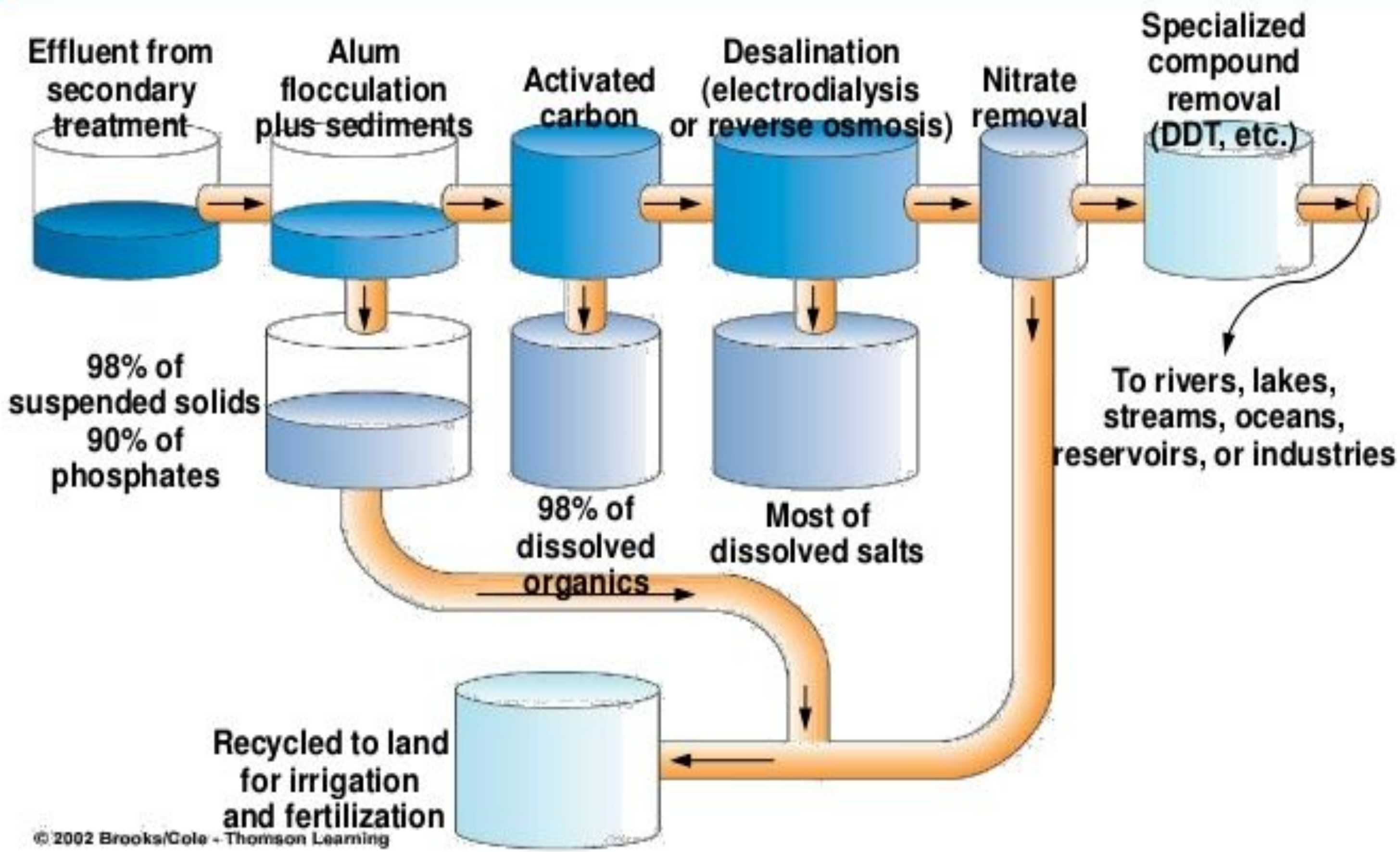


Source: Philadelphia Water Department

Primary and SecondaryTreatment

- pH Adjustment
- Flocculation
- Sedimentation
- Filtration
- Disinfection

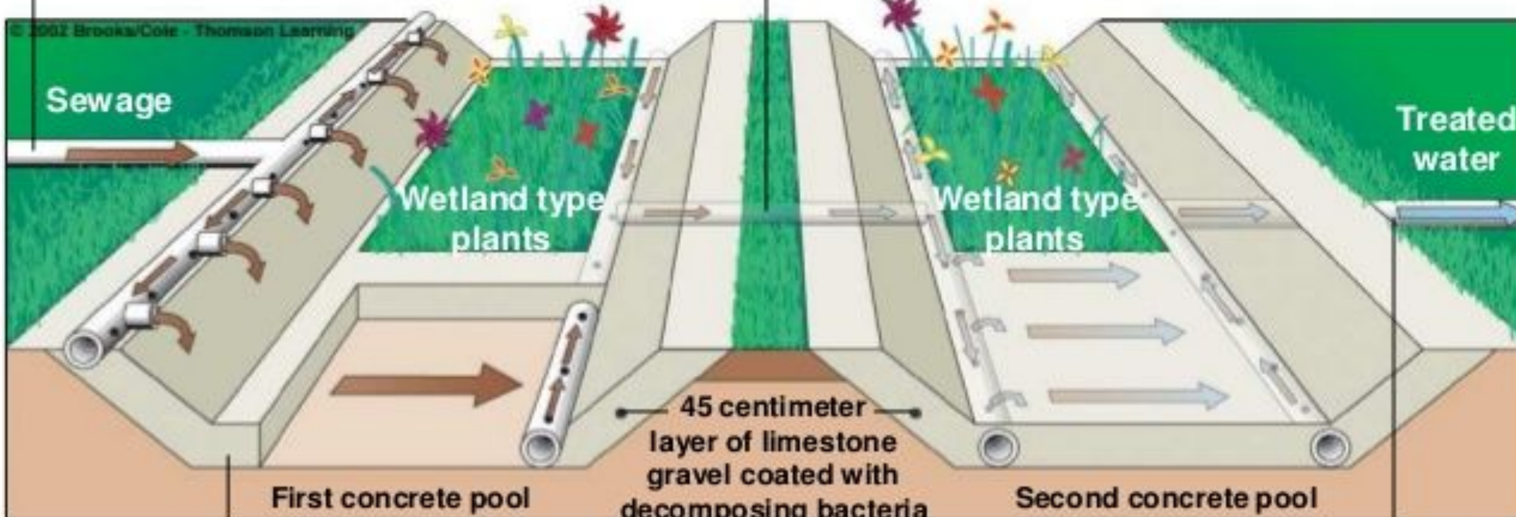
Tertiary Treatment



Wetland Addition

(1) Raw sewage drains by gravity into the first pool and flows through a long perforated PVC pipe into a bed of limestone gravel.

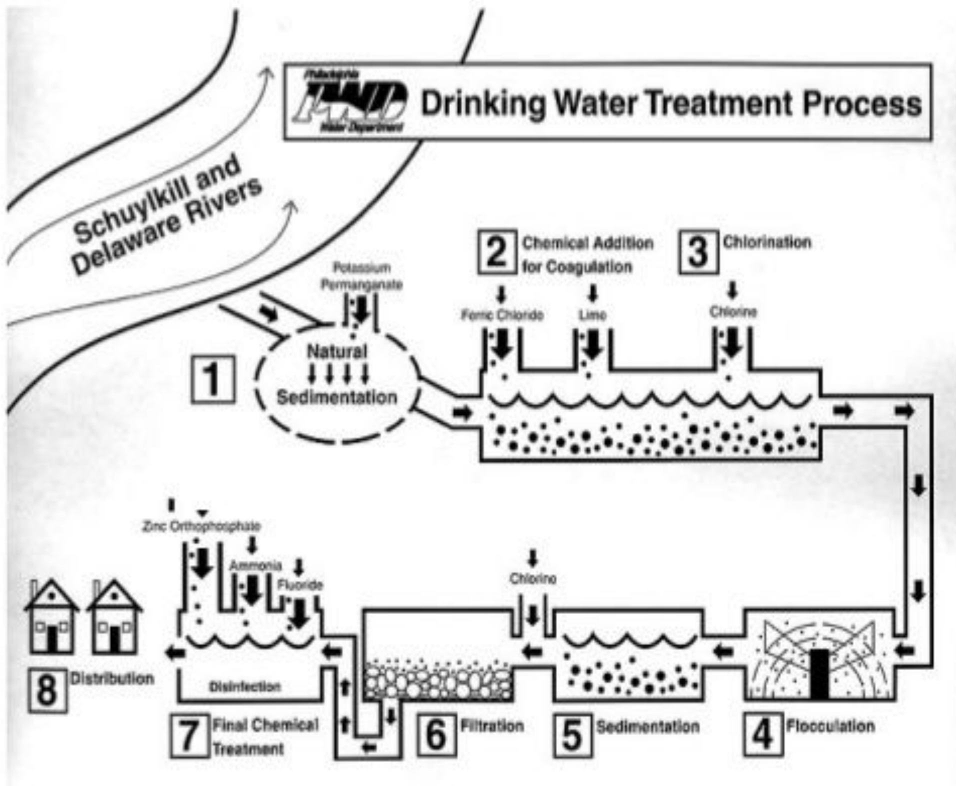
(3) Wastewater flows through another perforated pipe into a second pool, where the same process is repeated.



(2) Microbes in the limestone gravel break down the sewage into chemicals that can be absorbed by the plant roots, and the gravel absorbs phosphorus.

(4) Treated water flowing from the second pool is nearly free of bacteria and plant nutrients. Treated water can be recycled for irrigation and flushing toilets.

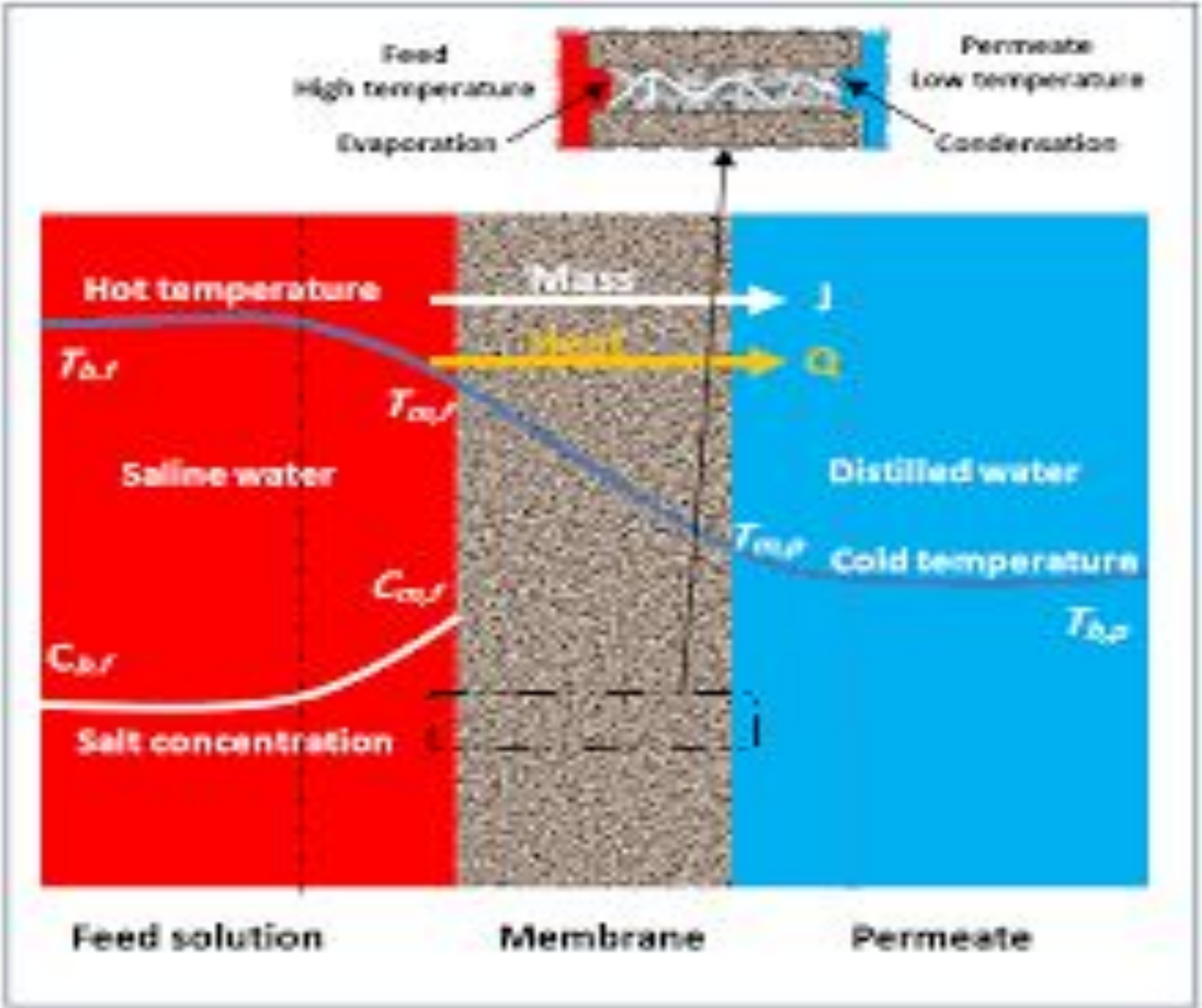
Drinking Water Treatment Process



Other Methods

- Boiling
- Activated Carbon
- Distillation
- Reverse Osmosis
- Direct Contact Membrane Distillation (DCMD)

- Direct Contact Membrane Distillation (DCMD) DCMD is the simplest MD configuration, in which a liquid phase (feed) with high temperature is in direct contact with the hot side of the membrane surface, and a cold aqueous phase is in direct contact with the permeate side. Therefore, volatile compounds evaporate at the hot liquid/vapor interface at the feed side. Having been passed through the membrane pores, the vapor phase will be condensed in the cold liquid/vapor interface at the permeate side.
- It is notable that the vapor pressure difference is induced by the temperature difference across the membrane and the hydrophobic nature of the membrane prevents the feed from penetrating through the membrane. Despite its simplicity, the conduction heat loss associated with this process is higher than in other configurations. Membrane modules in DCMD could be shell-and-tube or plate-and-frame employed under cross- flow or longitudinal flow.



Feed
High temperature
Evaporation
Permeate
Low temperature
Condensation
Hot temperature
Mass
J
Heat
Q
Saline water
Distilled water
Cold temperature
Salt concentration
Feed solution
Membrane
Permeate